

# Hajar Mrifag

Graduate Software Engineer | Backend & Full-stack | Class of 2027

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## PROFILE

Final-year Software Engineering student building React applications, Python and .NET APIs, and data-intensive systems. Project work spans database correctness, numerical optimization, automated testing and container deployment. Seeking graduate software engineering opportunities in 2027.

## EDUCATION

**Sichuan University** | Software Engineering | Expected graduation: 2027

## TECHNICAL SKILLS

**Languages:** Python, C#, TypeScript, SQL, C++ / CUDA

**Applications:** FastAPI, ASP.NET Core, React, PostgreSQL

**Delivery & data:** Docker, GitHub Actions, Git, PySpark, Hive, HDFS

## SELECTED ENGINEERING PROJECTS

### GridFlex | React, TypeScript, FastAPI, NumPy

- Built an energy-simulation workspace with a numerical engine, API, scenario comparison, forecasting, stress testing and optimization.
- Reduced the shift + battery pipeline from 71.731 to 2.246 ms (31.9x) on a recorded local 60-day benchmark; preserved numerical outputs for that case.
- Added bounded computation and caches, strict input validation and container deployment; release 0.2.0 includes 91 Python tests and five browser journeys.

### Aegis Claims | ASP.NET Core, PostgreSQL, React

- Developed a claims application with JWT authentication, API-enforced roles, document signature validation and an audit trail for status changes.
- Documented threat scenarios and deployment limits; added unit, integration and browser tests with automated security checks.

### Neobank API & Analytics | FastAPI, PostgreSQL, SQL, Docker

- Implemented atomic transfers, idempotent retries and consistently ordered account locks, with database-backed concurrency and rollback tests.
- Built reproducible analytics over 891,772 synthetic transactions, separating statistical uplift from incentive profitability.

### CUDA Transformer Kernels | CUDA C++, Nsight Compute

- Implemented and profiled GEMM and Softmax kernels with CPU-reference checks. Register-tiled FP32 GEMM reached 1.28 TFLOPS on a T4 at 1024 x 1024, 3.32x the naive kernel and 24% of cuBLAS (median of 20 runs).

## TEAM PROJECT

### SkyQuery - Backend & Data Engineering | University team project

- Worked on a flight-search and analysis platform using HDFS, Hive, PySpark and FastAPI over 15 million Expedia flight records.